Tim Beauchamp

Engineering Leadership and Applied AI

Belmont, California | tim@beauchampfamily.org | 650 333 1267
linkedin.com/in/tbeauchamp | github.com/timbeauchamp

Professional Profile

Engineering manager, hands-on developer, and inventor with a 35-year software career and more than 22 years at Adobe. Leads teams building AI and traditional software across native Mac, Windows, mobile, and complex web applications. Advances AI use throughout design, development, testing, deployment, and daily engineering management. Combines creative-product delivery with experience in communications, distributed systems, statistical software, and data platforms.

Interested in engineering leadership and technical partnerships in interoperable, distributed data communication, bringing practical experience in resilient applications, edge-to-cloud integration, telemetry, and product development.

Adobe

Engineering Manager / Computer Scientist | 2004–Present | San Jose, California

Experience spans a multi-platform mobile and desktop video editor, a mobile video-editing SDK, and a web-native video editor, with current focus on AI-enabled products and engineering practices.

AI Engineering and Product Direction

- Manage software developers who use AI extensively across the software lifecycle, delivering both AI and traditional applications across native and web platforms.
- Guide adoption of specification-driven agentic development alongside AI-assisted work on established codebases. Clarify requirements, engineering ownership, review expectations, tooling priorities, and quality assessment.
- Introduced a recurring learning forum to help engineers adapt to new platforms and agentic workflows, sharing practical successes, failures, and emerging techniques.
- Contribute technical direction to generative-media integration, including model execution boundaries, interactive preview, asset movement, and full-resolution output. Proposed reusable timeline operations within graph-based creative workflows.
- Involved in connecting AI tools and generative AI artifacts to traditional software through MCP. Prioritize generation telemetry, error reporting, and operational readiness.

People Leadership and Engineering Management

- Coach engineers toward broader ownership; provide specific performance feedback, support career growth and promotion, and make evaluation and calibration criteria visible.
- Guide teams through changing product charters and platforms while maintaining delivery. Recognized by colleagues for inclusive, respectful, collaborative team culture and welcoming new contributors.
- Apply AI to management through recurring communication digests, meeting action extraction, sprint consistency checks, and regression-retrospective preparation using connected engineering tools.

Adobe Contributions and Technical Leadership

Content Credentials and Cross Functional Delivery

- Coordinated Content Credentials delivery through rollout to all users of a web-native video editor in August 2026. Brought together client, media-service, export, QA, program, legal, and communications workstreams.
- Received explicit management recognition for pulling together workstreams and driving the Content Credentials initiative; engineering implementation remained a shared team achievement.
- Coordinate integration and release dependencies across Product, Design, engineering, quality, and service teams. Surface ownership, support, test coverage, and operational requirements before launch.

Performance Analysis and Engineering Tools

- Investigated media ingest using Splunk queries, Python analysis, and Claude assistance, grouping workloads by media characteristics to distinguish processing cost, concurrency, and queue contention.
- Analyzed upload traces to identify cached-block registration overhead, HTTP throttling, and Retry-After delays; proposed pacing, batching, and service-side investigation.
- Defined a performance-measurement framework spanning preparation, upload, processing, readiness, download, and display, with cache-state distinctions and workload normalization.
- Built a Python/FFmpeg generator for reproducible test media across durations, frame rates, bitrates, and file sizes. Built an internal blind listening-comparison service to evaluate audio time-stretch quality.
- Advocated architectural alignment between playback and export to address duplicate implementation and late-cycle delivery risk; brought client and service stakeholders into testing and dependency planning.

Cross Platform Products and Release Engineering

- Managed engineering for a professional nonlinear video editor spanning multiple platforms; maintained release delivery through mobile and desktop app stores and cloud services.
- Led a team developing an iOS and Android video-editing SDK; supported team preparation for a web-native video editor public beta.
- Diagnosed native build and release issues, including Jenkins/Python packaging incompatibility and iOS framework architecture blockers. Added stage/production environment switching to the native iOS video editor for testing.
- Organized innovation activities with project proposals, volunteer teams, check-ins, and demonstrations, creating opportunities for experimentation and shared learning.

Technical Breadth

AI application and integration: Claude, Cursor, ChatGPT, MCP, agentic workflows, and connected engineering tools. Systems and analysis: Python, SQL, Splunk, HTTP, APIs, media pipelines, benchmarking, and observability. Career experience also includes Java, Kotlin, Swift, JavaScript, C/C++, Objective-C, native mobile development, databases, and data warehousing.

Independent Engineering and Earlier Experience

Selected Independent Projects

Project Phaethon — Developed an edge-to-cloud sensing proof of concept combining camera nodes, irradiance measurements, cloud ingestion, archived observations, and operational dashboards. Implemented Python/FastAPI services and alert evaluation using precision/recall-related metrics; forecasting effectiveness remains under evaluation.

WhatPlaneIsThat — Developed Android and iOS aircraft-identification software integrating ADS-B data services, device position and orientation, geographic calculations, and camera-view overlays. Related work includes Android OpenDroneID parsing and orientation/astronomy libraries.

Network Wires — Developed connected-sensor firmware, AWS Lambda/DynamoDB services, and web and Android interfaces. Work spans hardware/software prototyping, IoT integration, and telemetry.

Local AI and automation — Built a Python gateway with task routing, structured results, timing, and explicit capability and uncertainty reporting. Integrated home automation software, MQTT, voice services, Linux-based embedded computing hosts, InfluxDB, and Grafana for local automation and troubleshooting.

Inventions in Communications and Distributed Software

Named co-inventor on 12 granted US software patents spanning real-time enterprise intelligence, event-driven collaboration, resilient networked applications, and adaptive user interfaces. Developed at Iteration/Istante Software; grants assigned to Oracle. Selected examples:

US 7,941,542 B2 — Maintaining application execution over an intermittent network connection.
US 8,402,095 B2 — Instant messaging collaboration.
US 8,001,185 B2 — Distributed rule evaluation in near real-time business intelligence.

Earlier Career

Istante Software — Founder / Senior Software Developer, real-time business intelligence; acquired by Oracle. AOL –Netscape — Senior Software Engineer, browser development. Data Digest — first engineer, statistical analysis software. Sagent — data warehousing development and QA. NetManage — TCP/IP and Internet application development and QA. Software Publishing Corporation — Harvard Graphics.

Additional technical experience includes Bayesian analysis and modeling software, document classification systems, database development, data warehousing, and 3D modeling and rendering.

Education and Additional Experience

Electronic Engineering — California Polytechnic State University, San Luis Obispo.
General Maintenance Mechanic — Antarctic Support Associates, Amundsen–Scott South Pole Station, Antarctica.